

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):	
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)	
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.	
ê by polarization plug into templates above	

Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$	
R refl. pwr	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$	
T trans. pwr	$ \tau_{\perp} ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	
R+T total	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$	

Notes: sin θ_t = √μ₁ε₁/μ₂ε₂ sin θ_i; nonmag. ⇒ η₂/η₁ = n₁/n₂.

Intrinsic impedance η_i plug into table above	
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$	$\eta_0 = 120\pi \approx 377 \Omega$
Default μ _r = 1 (air, dielectric). Use μ _r ≠ 1 only if problem says “ferromagnetic”/iron/ferrite/etc.	
Low-loss correction $\eta_c \sigma/(\omega\epsilon) \ll 1$	
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$	
For Γ/τ just use η ₀ /√ε _r . The j term is for η _c , θ _η . See LOSSY MEDIA — 5 CASES.	

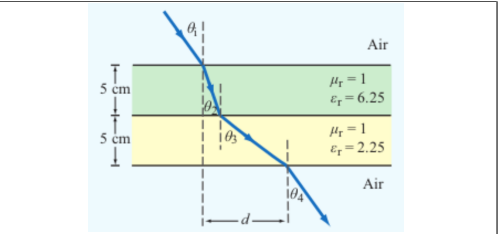
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	$\theta_{\text{exit}} = \theta_{\text{incident}}$
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
	$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$	
Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon'}{2\omega \epsilon_r}\right)$ α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ_η} see below
Good cond.	> 10 ²	η _c ≈ (1+j)√πfμ/σ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ _r = 1	
α ≈ $\frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)	
Exact (quasi-cond) X = √[1 + (ε''/ε') ²]	
α = ω√ $\frac{\mu \epsilon'}{2}$ √X-1, β = ω√ $\frac{\mu \epsilon'}{2}$ √X+1	
θ _η = $\frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$	TM mode $H_z = 0, E_z \neq 0$
Dimensions a × b with a > b along $\hat{x}, \hat{y}$; wave in $\hat{z}$.	
<i>E_x form (TE)</i> read m, n from arguments	
$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$	
<i>coef. on x = mπ/a; coef. on y = nπ/b.</i>	

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

e^{-jβz} factor omitted. k_c² = (mπ/a)² + (nπ/b)². Ê in V/m, Ĥ in A/m, Z in Ω.

TE mode <i>generator: Ê_z</i>	$\hat{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$
$\hat{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\hat{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\hat{E}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TE}, \hat{H}_y = \hat{E}_x/Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c/f)^2}$	

TM mode <i>generator: Ê_z</i>	$\hat{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$
$\hat{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\hat{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\hat{H}_z = 0, \hat{H}_x = -\hat{E}_y/Z_{TM}, \hat{H}_y = \hat{E}_x/Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$	

TEM plane wave *for reference*
 $\hat{E}_x = E_0 e^{-j\beta z}, \hat{H}_y = \hat{E}_x/\eta$

$$\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega\sqrt{\mu\epsilon}$$

Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k\sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)	
<i>Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10⁸ (m/s), m, n (unitless).</i>	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s), } u_p u_g = u_{p0}^2$	
t = L/u _g (s) travel time through guide of length L	
<i>Higher modes have lower u_g ⇒ arrive later.</i>	

BOUNDARY

η _i , η _t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = Γ ²
k _i = ω√ε _{r,i} /c (rad/m)
α ₂ (Np/m), β ₂ (rad/m): lossy
z = 0 — boundary plane (m)
η ₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ _i , θ _t — angles from normal
n _i = √ε _{r,i} — index of refr.
t _i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: E ⊥ this plane
: E this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f _{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z _{TE} /TM — wave imp. (Ω)
E _x — E-field comp. (V/m)
H _y — H-field comp. (A/m)
η = η ₀ /√ε _r

VELOCITIES

u _{p0} = c/√ε _r — bulk phase vel.
u _p = u _{p0} /√[1 - (f _c /f) ²]
u _g = u _{p0} √[1 - (f _c /f) ²]
u _p u _g = u _{p0} ²
t = L/u _g — travel time
c = 3 × 10 ⁸ m/s

POWER & UNITS

%P _r , %P _t — pwr refl./trans. (unitless)
%P _r = 100 Γ ²
%P _t = 100 τ ² η ₁ /η ₂
%P _r + %P _t = 100
σ (S/m), ε (F/m), μ (H/m)
ε ₀ = 8.85 × 10 ⁻¹² F/m
μ ₀ = 4π × 10 ⁻⁷ H/m

